



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

Practical No.2

Implement Stack with insertion, deletion, traversal operations (CI and GUI)

Aim: Write a program to implement Stack with insertion, deletion, traversal operations

- Implement push, pop, peek using arrays or linked lists.
- Solve problems like delimiter matching or undo mechanism.
- Convert expressions from prefix to postfix and evaluate them.

Command Line

Code:-

#S104 Pratik Rajbhar

stack = []

undo_stack = []

def push():

 item = input("Enter element: ")

 stack.append(item)

 undo_stack.append(("push", item))

 print(item, "inserted successfully.")

def pop():

 if len(stack) == 0:

 print("Stack Underflow")

 else:

 item = stack.pop()

 undo_stack.append(("pop", item))

 print("Deleted Element:", item)

Roll No:-S104

Name:- Pratik Rajbhar



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

```
def peek():
```

```
    if len(stack) == 0:
```

```
        print("Stack is Empty")
```

```
    else:
```

```
        print("Top Element:", stack[-1])
```

```
def display():
```

```
    if len(stack) == 0:
```

```
        print("Stack is Empty")
```

```
    else:
```

```
        print("\nStack Elements (Top to Bottom)")
```

```
        for item in reversed(stack):
```

```
            print(item)
```

```
# Delimiter Matching
```

```
def delimiter_matching():
```

```
    expression = input("Enter Expression: ")
```

```
    temp = []
```

```
    pairs = {'(': ')', '[': ']', '{': '}'}
```

```
    for ch in expression:
```

```
        if ch in "({[":
```

```
            temp.append(ch)
```

```
        elif ch in ")}]":
```

```
            if not temp or temp.pop() != pairs[ch]:
```

```
                print("Not Balanced")
```

```
    return
```

Roll No:-S104

Name:- Pratik Rajbhar



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

```
if len(temp) == 0:
```

```
    print("Balanced Expression")
```

```
else:
```

```
    print("Not Balanced")
```

```
# Undo
```

```
def undo():
```

```
    if len(undo_stack) == 0:
```

```
        print("Nothing to Undo")
```

```
        return
```

```
    action = undo_stack.pop()
```

```
    if action[0] == "pop":
```

```
        if stack:
```

```
            stack.pop()
```

```
        elif action[0] == "push":
```

```
            stack.append(action[1])
```

```
    print("Undo Successful")
```

```
# Prefix to Postfix
```

```
def prefix_to_postfix():
```

```
    prefix = input("Enter Prefix Expression: ")
```

```
    temp = []
```

```
    for ch in reversed(prefix):
```

```
        if ch.isalnum():
```

```
            temp.append(ch)
```

```
        else:
```

```
            op1 = temp.pop()
```

Roll No:-S104

Name:- Pratik Rajbhar



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

```
op2 = temp.pop()
```

```
temp.append(op1 + op2 + ch)
```

```
print("Postfix Expression:", temp[-1])
```

```
# Evaluate Postfix
```

```
def evaluate_postfix():
```

```
    postfix = input("Enter Postfix Expression: ")
```

```
    temp = []
```

```
    for ch in postfix:
```

```
        if ch.isdigit():
```

```
            temp.append(int(ch))
```

```
        else:
```

```
            b = temp.pop()
```

```
            a = temp.pop()
```

```
            if ch == '+':
```

```
                temp.append(a+b)
```

```
            elif ch == '-':
```

```
                temp.append(a-b)
```

```
            elif ch == '*':
```

```
                temp.append(a*b)
```

```
            elif ch == '/':
```

```
                temp.append(a//b)
```

```
            elif ch == '^':
```

```
                temp.append(a**b)
```

```
    print("Result =", temp.pop())
```

```
# Menu
```

Roll No:-S104

Name:- Pratik Rajbhar



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

while True:

```
print("\n===== STACK MENU =====")
```

```
print("1. Push")
```

```
print("2. Pop")
```

```
print("3. Peek")
```

```
print("4. Display")
```

```
print("5. Delimiter Matching")
```

```
print("6. Undo")
```

```
print("7. Prefix to Postfix")
```

```
print("8. Evaluate Postfix")
```

```
print("9. Exit")
```

```
choice = input("Enter Choice: ")
```

```
if choice == '1':
```

```
    push()
```

```
elif choice == '2':
```

```
    pop()
```

```
elif choice == '3':
```

```
    peek()
```

```
elif choice == '4':
```

```
    display()
```

```
elif choice == '5':
```

```
    delimiter_matching()
```

```
elif choice == '6':
```

```
    undo()
```

```
elif choice == '7':
```

Roll No:-S104

Name:- Pratik Rajbhar



Laxmi Charitable Trust's Sheth L.U.J College of Arts & Sir M.V. College Of Science & Commerce

```
File Edit Shell Debug Options Window Help
2. Pop
3. Peek
4. Display
5. Delimiter Matching
6. Undo
7. Prefix to Postfix
8. Evaluate Postfix
9. Exit
Enter Choice: 2
Deleted Element: 256

===== STACK MENU =====
1. Push
2. Pop
3. Peek
4. Display
5. Delimiter Matching
6. Undo
7. Prefix to Postfix
8. Evaluate Postfix
9. Exit
Enter Choice: 3
Top Element: 2

===== STACK MENU =====
1. Push
2. Pop
3. Peek
4. Display
5. Delimiter Matching
6. Undo
7. Prefix to Postfix
8. Evaluate Postfix
9. Exit
Enter Choice: 5
Enter Expression: (A + B) * C
Balanced Expression

===== STACK MENU =====
1. Push
2. Pop
3. Peek
4. Display
5. Delimiter Matching
6. Undo
7. Prefix to Postfix
8. Evaluate Postfix
9. Exit
Enter Choice: 6
Undo Successful

===== STACK MENU =====
1. Push
2. Pop
3. Peek
4. Display
5. Delimiter Matching
6. Undo
7. Prefix to Postfix
8. Evaluate Postfix
9. Exit
Enter Choice: |
```



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

GUI

```
import tkinter as tk
```

```
from tkinter import messagebox
```

```
stack = []
```

```
# Push
```

```
def push():
```

```
    item = entry.get()
```

```
    if item == "":
```

```
        messagebox.showwarning("Warning", "Enter a value")
```

```
        return
```

```
    stack.append(item)
```

```
    entry.delete(0, tk.END)
```

```
    display()
```

```
# Pop
```

```
def pop():
```

```
    if len(stack) == 0:
```

```
        messagebox.showerror("Error", "Stack Underflow")
```

```
    else:
```

```
        messagebox.showinfo("Deleted", stack.pop())
```

Roll No:-S104

Name:- Pratik Rajbhar



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

display()

Peek

def peek():

if len(stack) == 0:

messagebox.showinfo("Stack", "Stack Empty")

else:

messagebox.showinfo("Top Element", stack[-1])

Display

def display():

listbox.delete(0, tk.END)

for item in reversed(stack):

listbox.insert(tk.END, item)

Delimiter Matching

def delimiter():

expression = entry.get()

temp = []

pairs = {'(': ')', '[': ']', '{': '}'}

for ch in expression:

Roll No:-S104

Name:- Pratik Rajbhar



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

```
if ch in "({":
```

```
    temp.append(ch)
```

```
elif ch in ")}":
```

```
    if not temp or temp.pop() != pairs[ch]:
```

```
        messagebox.showinfo("Result", "Not Balanced")
```

```
    return
```

```
if len(temp) == 0:
```

```
    messagebox.showinfo("Result", "Balanced")
```

```
else:
```

```
    messagebox.showinfo("Result", "Not Balanced")
```

```
# Prefix to Postfix
```

```
def prefix_postfix():
```

```
    prefix = entry.get()
```

```
    temp = []
```

```
    for ch in reversed(prefix):
```

Roll No:-S104

Name:- Pratik Rajbhar



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

```
if ch.isalnum():
```

```
    temp.append(ch)
```

```
else:
```

```
    op1 = temp.pop()
```

```
    op2 = temp.pop()
```

```
    temp.append(op1 + op2 + ch)
```

```
messagebox.showinfo("Postfix", temp[-1])
```

```
# Evaluate Postfix
```

```
def evaluate():
```

```
    postfix = entry.get()
```

```
    temp = []
```

```
    for ch in postfix:
```

```
        if ch.isdigit():
```

```
            temp.append(int(ch))
```

```
        else:
```

Roll No:-S104

Name:- Pratik Rajbhar



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

```
b = temp.pop()
```

```
a = temp.pop()
```

```
if ch == '+':
```

```
    temp.append(a+b)
```

```
elif ch == '-':
```

```
    temp.append(a-b)
```

```
elif ch == '*':
```

```
    temp.append(a*b)
```

```
elif ch == '/':
```

```
    temp.append(a/b)
```

```
elif ch == '^':
```

```
    temp.append(a**b)
```

```
messagebox.showinfo("Result", temp.pop())
```

```
# Window
```

```
root = tk.Tk()
```

Roll No:-S104

Name:- Pratik Rajbhar



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

```
root.title("Stack Implementation")
```

```
root.geometry("350x500")
```

```
tk.Label(root,text="Enter Value / Expression",font=("Arial",12)).pack()
```

```
entry = tk.Entry(root,width=30)
```

```
entry.pack(pady=5)
```

```
tk.Button(root,text="Push",command=push).pack(fill="x")
```

```
tk.Button(root,text="Pop",command=pop).pack(fill="x")
```

```
tk.Button(root,text="Peek",command=peek).pack(fill="x")
```

```
tk.Button(root,text="Delimiter Matching",command=delimiter).pack(fill="x")
```

```
tk.Button(root,text="Prefix to Postfix",command=prefix_postfix).pack(fill="x")
```

```
tk.Button(root,text="Evaluate Postfix",command=evaluate).pack(fill="x")
```

```
listbox = tk.Listbox(root,height=10)
```

```
listbox.pack(fill="both",pady=10)
```

```
root.mainloop()
```



Laxmi Charitable Trust's Sheth L.U.J College of Arts & Sir M.V. College Of Science & Commerce

Stack Implementation

Enter Value / Expression

Push

Pop

Peek

Delimiter Matching

Prefix to Postfix

Evaluate Postfix

42
5
23
2
5



Stack Implementation

Enter Value / Expression

Push

Pop

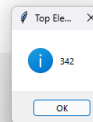
Peek

Delimiter Matching

Prefix to Postfix

Evaluate Postfix

342
45
523
42
3
45
5
5
5
5





Laxmi Charitable Trust's Sheth L.U.J College of Arts & Sir M.V. College Of Science & Commerce

```
"IDLE Shell 3.14.6"
File Edit Shell Debug Options Window Help

4. Display
5. Delimiter Matching
6. Undo
7. Prefix to Postfix
8. Evaluate Postfix
9. Exit
Enter Choice: 1
Enter element: 2
2 inserted successfully.

===== STACK MENU =====
1. Push
2. Pop
3. Peek
4. Display
5. Delimiter Matching
6. Undo
7. Prefix to Postfix
8. Evaluate Postfix
9. Exit
Enter Choice: 1
Enter element: 256
256 inserted successfully.

===== STACK MENU =====
1. Push
2. Pop
3. Peek
4. Display
5. Delimiter Matching
6. Undo
7. Prefix to Postfix
8. Evaluate Postfix
9. Exit
Enter Choice: 2
Deleted Element: 256
```

Stack Implementation

Enter Value / Expression

ABC/AK**

Push

Pop

Peek

Delimiter Matching

Prefix to Postfix

Evaluate Postfix

342

45

523

42

3

45

5

5

5

5